

**RAJIV GANDHI COLLEGE OF ENGINEERING AND
TECHNOLOGY**

COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT - 1

Prepared by: MOHAMED B SIRAJUDEEN

Reg No : 22TD0053

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Unit : I

Teacher's Sign :

Department Vision and Mission

Vision

Cultivate student as technocrats, teachers and entrepreneurs in the field of computer science and engineering.

Mission.

- * To impart quality education in Computer Science and Engineering by means of learning techniques and value-added courses.
- * To inculcate professional excellence and research culture by encourage projects in cutting-edge technologies through industry interaction.
- * To build leadership skills, ethical values and teamwork among the students.
- * To strengthen the collaboration of department and industry through internships, mentorships and professional body activities.

i) Research and summarize the design philosophy of ARM architecture.

The design philosophy of ARM architecture focuses on low power consumption, high code density, and low cost. ARM processors are used in a wide range of devices, including smart phones, laptops and tablets.

* **Simplicity and Efficiency.**

→ Reduced Instruction set computing (RISC)

ARM's architecture adheres to the RISC design principles, which focus on simplifying the instruction set. Unlike complex instruction set computing (CISC) architecture, RISC architecture use a smaller, more streamlined set of instructions, each performing simple operations. This allows for:

* **Faster Execution.**

This reduced instruction set results in simpler decoding and faster execution of instructions, allowing for higher clock speeds.

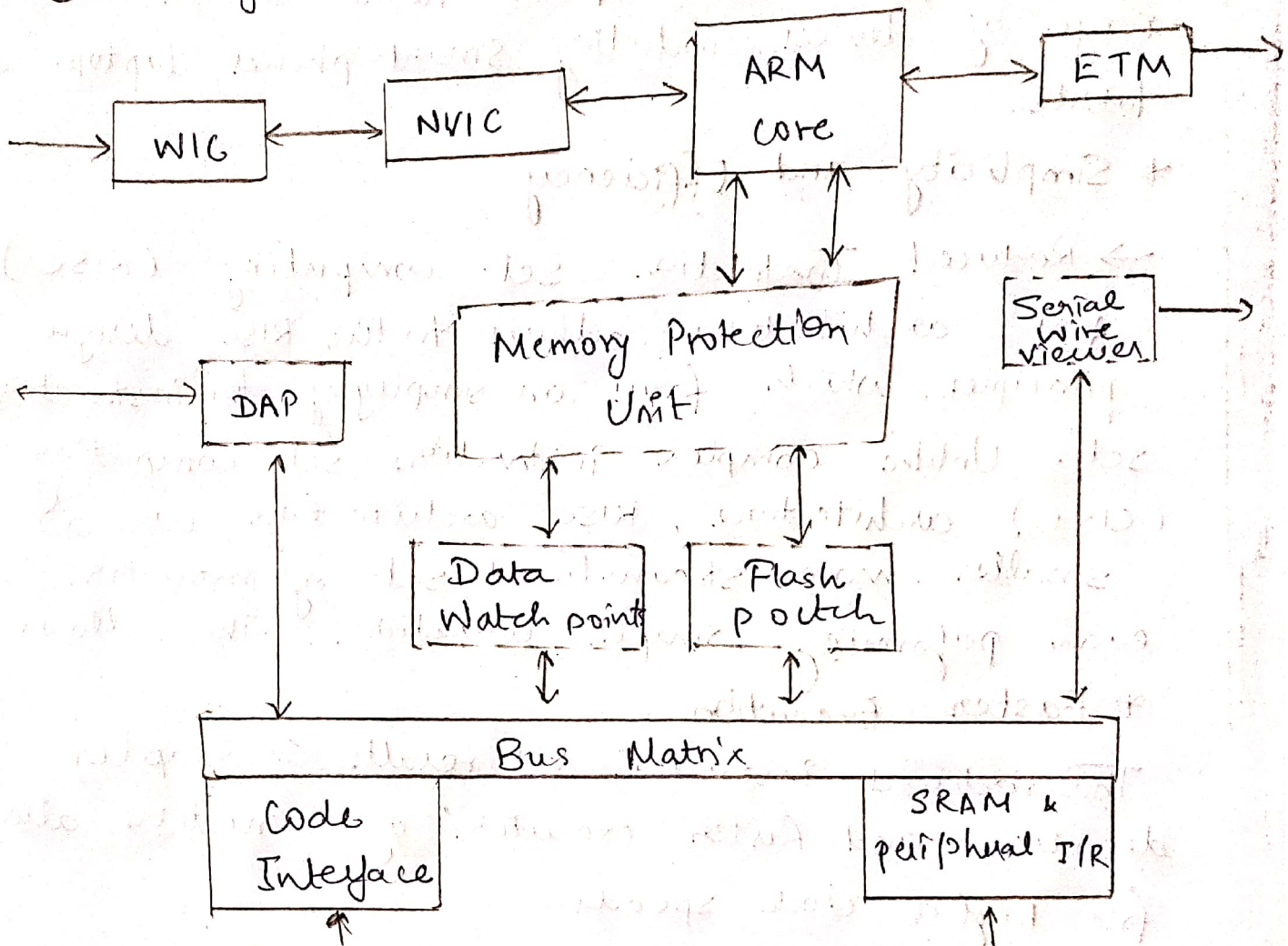
* **Efficient pipeline.**

With fewer instructions, ARM processors are optimized for pipelining, enabling more efficient execution of instructions in parallel. This contributes to reduced cycles per instruction (CPI)

* **Predictable performance.**

Fewer, simpler instructions make ARM processors more predictable in terms of performance, a key advantage in real-time computing and embedded systems.

Smaller Instruction set
 ARM instructions are typically fixed-length, further simplifying decoding and encoding better pipelining. Most instructions execute in a single clock cycle, which enhances performance.



ARM processor and its Features

* Power Efficiency

Low power consumption.

ARM processors are renowned for their power efficiency, a crucial feature in mobile, embedded, and battery-powered devices like smartphones, tablets, IoT devices, and wearables.

Simplicity of design:

The simpler instruction set requires fewer transistors to process instructions, resulting in lower overall power consumption.

* **Energy efficiency:** ARM processors can deliver high performance per watt, meaning that they can perform tasks effectively without draining much power, making them ideal for devices with strict power budgets.

* Scalability

Wide Range of Application:

ARM's scalability allows it to be used in a broad spectrum of devices, from simple microcontrollers to complex multicore processors used in high performance servers. ARM achieves this scalability through

* Flexible core Designs.

ARM licenses various core designs, from simple low-power cores for embedded systems to high-performance cores for servers and desktops. This flexibility allows ARM to target a wide range of performance requirements.

* Multi-core configuration.

ARM processors support multi-core architecture, making them suitable for everything from low-power smartphones to high-performance computing environments.

→ Customizable Licensing model

ARM's business model involves licensing its processor designs to third-party manufacturers, which allows those manufacturers to tailor ARM cores for specific applications.

→ Load store Architecture

→ Memory Access via Registers.

ARM follows a load-store architecture, meaning that instructions can only access memory via specific load or store operations.

* Faster Execution,

Since the CPU does not have to interact with memory for most operations, it avoids delays associated with memory access.

* Simplified Instruction set.

The load-store model simplifies the CPU's design, contributing to ARM's overall efficiency. The processor only needs to manage register-to-register operations and load/store interactions with memory.

* Enhanced Multicore and parallelism

Multicore Support

ARM processors have increasingly incorporated multi-core designs, which enable parallel processing. This is crucial for modern computing, where workloads are distributed across multiple cores to improve overall performance.

* Symmetric Multiprocessing (SMP)

ARM supports SMP configurations, which means multiple cores can operate independently but share the same memory space. This feature is crucial for scaling ARM processors for demanding applications, from smartphones to high performance computing systems.

* Extensibility and Licensing

→ Licensing Model

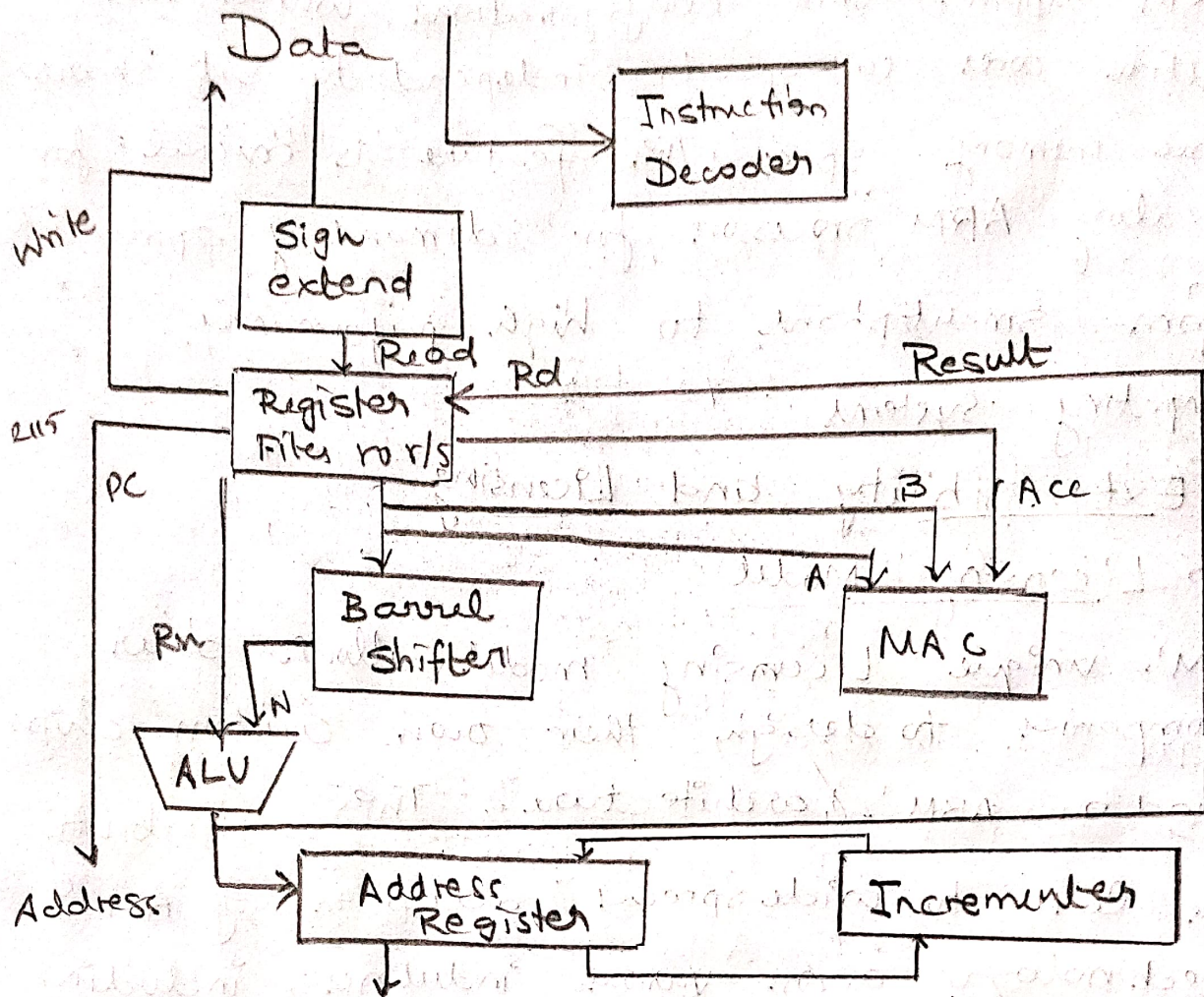
ARM's unique licensing model allows other companies to design their own custom chips based on ARM's architecture. This flexibility has led to widespread adoption of ARM technology across various industries, including consumer electronics and industrial applications.

* Custom Cores

Manufacturers can design custom ARM cores tailored to specific needs, providing an additional level of optimization ~~and~~ and flexibility. This customization allows ARM to penetrate diverse markets with varying requirements for performance, power consumption and cost.

ARM's design philosophy centers around creating processors that are simple, energy-efficient and scalable while maintaining flexibility for a broad range of applications. Its emphasis on a reduced instruction set, low power consumption, scalability, and licensing model has made ARM

One of the most successful and widely used architectures.

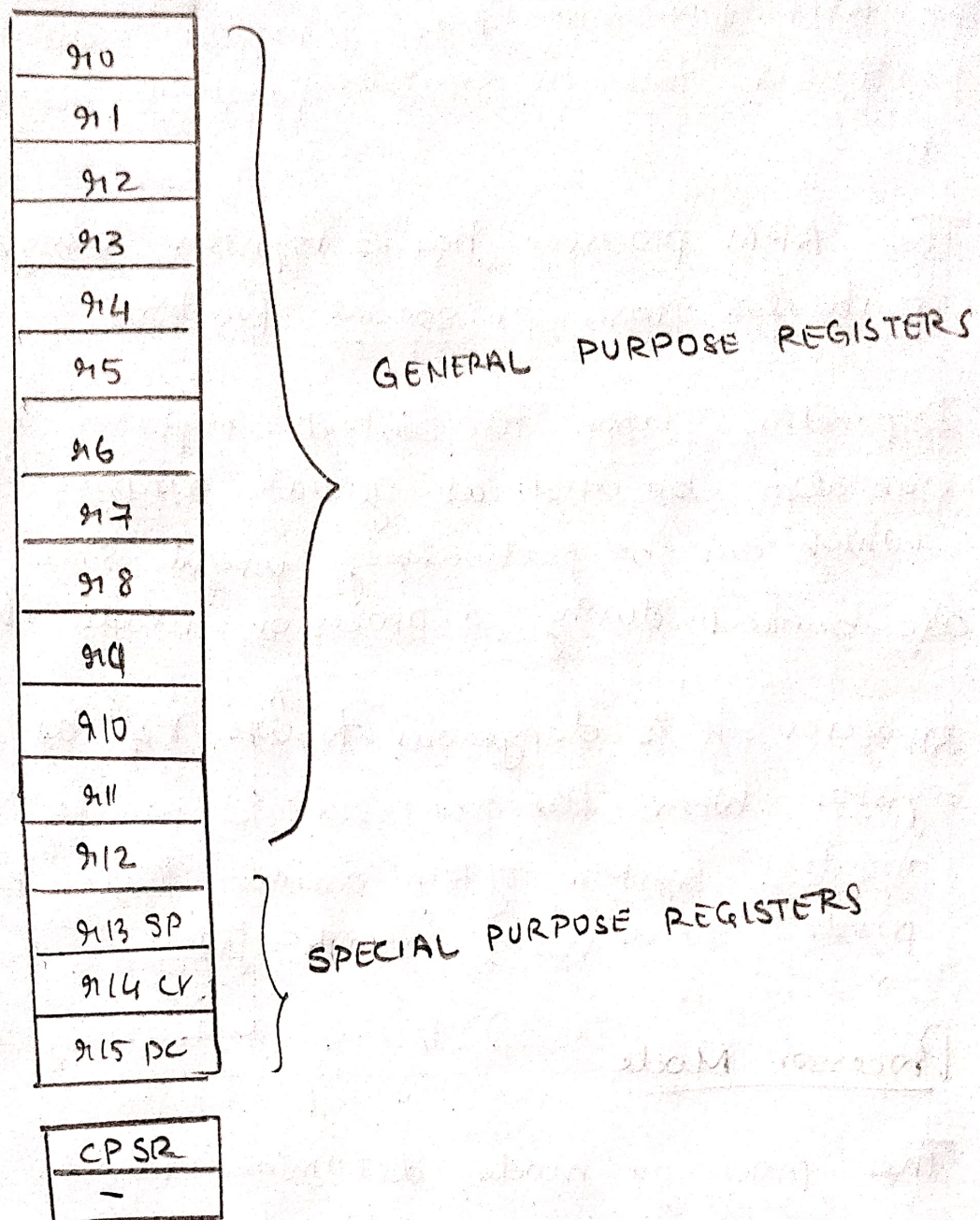


Data enters a processor code through the data and the data may be an instruction to execute or a load.

The data items are placed in the registers storage basic bank made up of 32-bit registers.

2) Describe the different types of registers used in ARM processor, including general-purpose and special purpose registers.

General purpose registers hold either data or an address. They are identified with the letter r prefixed to the register number. For ex: register 4 is given the value r_4 .



The figure shows the active registers available in user mode - a protected mode normally used when executing applications.

The processor can operate in seven different modes

All the registers shown are 32 bit in size

There are up to 18 active registers : 16 data registers and 2 processor status registers.

The data registers are visible to the programmer as R0 to R15

The ARM processor has 3 registers assigned to a particular task or special functions: r_{13} , r_{14} and r_{15} .

The ARM processor has 3 registers assigned to a particular task or special functions.

Depending upon the context, registers r_{13} and r_{14} can also be used as general purpose registers which can be particularly useful since these registers are banked during a processor mode change.

However, it is dangerous to use r_{13} as a general register. When the processor is running any form of operating system often assume that r_{13} always points to a valid stack frame.

Processor Mode

The processor mode determines which registers are active and the access rights to the CPSR registers itself.

Each processor mode is either privileged or non-privileged.

Undefined mode is used when the processor encounters an instruction that is undefined or supported by the implementation. User mode is for programs and applications.

3) What are the different types of interrupts in embedded systems?

An interrupt is a signal to the processor emitted by hardware or software indicating an event that needs immediate attention.

When ever an interrupt occurs, the controller completes the execution of the current instruction Routine (ISR) or Interrupt handler).

In Embedded Systems, interrupt handling is essential for managing asynchronous events and ensuring real-time responsiveness.

They are categorized based on their source, behaviour and handling mechanisms.

The types of interrupts commonly encountered in embedded systems are:

1) Hardware Interrupts

→ Generated by external devices or on-chip peripherals to signal events.

* Example:

External: Button press, sensor input, UART data reception.

Internal: Timer overflow, ADC conversion complete, Watchdog timer reset.

Usage: Handle real time events (eg: reading sensor data, time critical tasks)

2) Software Interrupts:

Triggered intentionally by software via specific instruction
(eg. SWI in ARM, INT in x86)

• Example

• System calls (eg: OS tasks)

• Debugging breakpoints.

→ Usage: Facilitate controlled transitions to kernel mode or service routines.

3) Maskable Interrupts.

In interrupt handling, it can be enabled, disabled via software (eg: using `sei` [enable] or `cli` [disable] in AVR)

• Examples: Most hardware interrupts (timers, GPIO)

→ Handling: controlled by the Global Interrupt Enable (GIE) flag.

4) Non-Maskable Interrupt (NMI)

• Cannot be disabled and must always be serviced immediately.

Examples: Critical failures (power loss, hardware faults, watch dog timer resets)

→ Usage: Handle emergencies requiring guaranteed CPU attention.

5) Fixed priority Interrupts.

• Predefined priority levels.

→ Behaviour: Higher priority interrupts preempt lower priority ones.

6) Non-prioritised (polled) Interrupts:

All interrupts share the same priority.
Handling - processed in the order they occur.

7) Edge-Triggered Interrupts.

Activated by a transition (rising / falling edge) on a signal.

Example: Button presses, and pulse detection.

Advantage: Avoids repeated triggering during the signal hold.

8) Level-Triggered Interrupt Handling.

Activated by a sustained signal level (high/low)

Examples: UART data ready, continuous sensor alert

Risk: May retrigger if the interrupt flag is not cleared.

4) Discuss the target Application for each ARM processor family.

ARM PROCESSOR FAMILIES.

ARM has designed a number of processors that are grouped into different families according to the core they are.

The families are based on the ARM 7, ARM 9, ARM 10 and ARM 11 cores.

The postfix numbers 7, 9, 10 and 11 indicate core designs.

The ascending number equates to an increase in performance and sophistication.

ARM8 was developed but was soon superseded.

Within each ARM family, there are a number of variations of memory management, cache and TCM processor extensions.

ARM 7 Family

The ARM7 core has a von Neumann style architecture, where data and instructions use the same bus.

The core has a three-stage pipeline and executes the architecture ARMv4T instruction set.

The ARM7 TDMI was the first of a new range of processor introduced in 1995 by ARM.

It is currently ~~in 1995~~ very popular core and is used in many 32-bit embedded processors. It provides a very good performance to power ratio.

ARM 9 Family

The ARM9 family was announced in 1997.

Because of its five stage pipeline, the ARM9 processor can run at higher clock frequencies than the ARM7 family.

The extra stages improve the overall performance of the processor.

The memory system has been redesigned to follow the Harvard architecture, which separates the data and instruction buses.

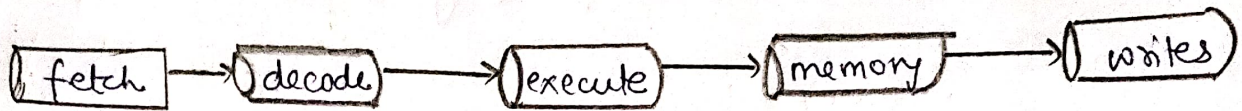
The next processor in the ARM9 Family were based on the ARM9E-S core. This core is a synthesizable

version of the ARM 9E-S core.

There are two variations: the ARM 946E-S and the ARM 966E-S.

Both execute architecture VSE instruction.

This is important when debugging application with time-critical segments.



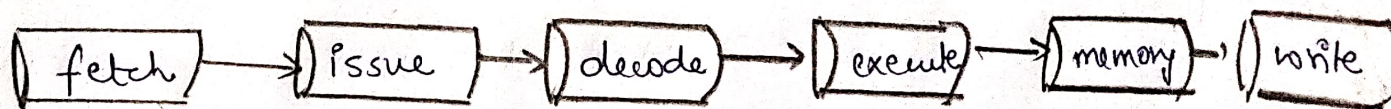
ARM10 Family

The ARM10 announced in 1999, was designed for performance.

It extends the ARM9 pipeline to six stages. It also supports an optional vector floating point (VFP) Unit, which adds a seventh stage to the ARM10 pipeline.

The ARM1020E is the first processor to use an ARM10E core like the ARM9E, it includes the enhanced E instruction.

This processor has the performance of the ARM10 with the flexibility of an ARM926EJ-S.



ARM11 Family

The ARM11 365-S announced in 2003, was designed for high-performance and power efficient application.

ARM1136-JS was the first processor implementation to execute architecture ARMv6 instruction.

It incorporate an eight - stage pipeline with
separate load store and arithmetic pipelines.
Included in the ARMV6 instruction are
single instruction multiple Data (SIMD)
extension for med processing, specifying designed
to increase video processing performance.

Discuss the target application for each ARM processor. ARM processors are widely used in various applications due to their energy efficiency, scalability and flexibility. ARM designs different processor families, each optimized for specific workloads.

1. ARM Cortex-A series - Application Processors

- These processors are designed for high performance computing, supporting full operating systems like Android, Linux and Windows on ARM. They feature multi-core architectures, out-of-order execution, and advanced power management, making them ideal for consumer electronics and computing devices.

2. ARM Cortex-R series - Real-Time Processors (Safety-Critical System)

- Cortex-R processors focus on real time, deterministic performance with low-latency execution. They are used in safety-critical applications where response time and reliability are crucial, such as embedded systems in vehicles and robotic control.

3. ARM Cortex-M series - Microcontrollers (Low power Embedded System)

Cortex-M processors are designed for low-power, cost sensitive applications. They have a simplified architecture with integrated peripherals, making them suitable for microcontrollers in IoT and consumer electronics.

7. ARM Neoverse - Server and cloud computing

Neoverse processors are optimised for high-performance computing in data centers.

Companies like Amazon and Ampere use Neoverse designs to build energy-efficient cloud servers that handle massive workloads.

5. ARM SecurCore - Security and Cryptographic Applications

These processors are designed for hardware-level security, features cryptographic accelerators and tamper-resistant features, making them ideal for secure transactions and identity verification.

6. ARM Ethos - AI and Machine Learning

ARM Ethos processors are optimized for deep learning and neural network acceleration, enabling AI-powered applications with minimal power consumption.

Each ARM processor is designed to meet specific performance, power, and security requirements.

From high-performance computing to low power embedded systems and real-time applications.